

France-based Helite has developed hip airbags, which protect wearers' bones from the impact of a fall. A Japanese firm called Xenoma also helps reduce the risk of falls by adapting its smart clothing products initially designed for golfers. Products like these help elderly and disabled people to lead an independent life without always expecting someone's help even to move around. Not just personal safety but their confidence also gets a boost from such products.

Rise of 'wearapeutics.' A new generation of smart, connected, sensor-based wearable medical devices is making healthcare much more efficient. These devices continuously monitor the patient's condition, give reminders about medicines to be taken, or alert the medical practitioner or caregiver when the situation requires attention.

Insulet's OmniPod insulin patch pump, Proteus' smart pills that send an alert when medicines are taken or skipped, Unilife's wearable injectors, Vivalnk's eSkin sensor patches and TempTraQ's comfortable 24-hour intelligent thermometer patches are some of the recent products. Patches and wearables are becoming increasingly comfortable, thanks to the development of biocompatible materials, some of which are made of stretchable semiconductor technologies such as elastomer-polymer. To avoid privacy and security mishaps, doctors and patients must check whether the devices they use comply



These hip airbags protect your bones from the impact of falls (Source: Helite)

with the Health Insurance Portability and Accountability Act (HIPAA).

Stay beautiful. At Consumer Electronics Show (CES) this year, L'Oréal unveiled My UV Patch, a next-generation wearable ultraviolet (UV) sensor. The tiny, stretchable patch can be stuck on your skin, bag, sunglasses or nails. It measures UV data and sends it to a smartphone app using near-field communications (NFC). Apart from collating the UV exposure and temperature from the patch, the smartphone app has an application programming interface (API) that combines information about local air quality to give you a complete picture. You can use this app to track your run or morning exercise too. Using all this info, you can fine-tune your beauty regime.

Get your beauty sleep

too. There are many smartwatches and fitness bands that monitor your sleep patterns and offer suggestions. Smart pillows and smart beds are also sighted here and there. Phillips recently launched a wearable device that not only monitors but also improves the quality of your sleep. The soft foam headband delivers customised tones to make you sleep deeply. Its pair of built-in sensors connect to



Might not sound like a lullaby, but white noise from Philip's headband helps you sleep deeper

your forehead and behind your ears to monitor brain activity. As soon as it detects a condition of deep sleep or slow-wave sleep, it starts playing a white noise in a slowly repeating pattern to help you stay in that state of deep sleep. White noise is a sound that is consistent across all audible frequencies. It helps to mask environmental sounds, helping you sleep better.

It is evident that wearables have broken all barriers and gotten into almost every field. There are connected safety jackets for miners and deep-divers, uniforms with close combat sensors for army men, and GPS-enabled patches that let you keep an eye on your kids or elderly parents who need care. Name the need, and there is an existing or upcoming wearable for it. Indeed, this field is growing rapidly and the best part is that there is more room to grow with so many opportunities everywhere we look! **EFY**



L'Oréal's My UV Patch lets you learn more about your UV exposure